Project Notes: Assumptions, Limitations and Corrections

ShopSmart Retail Ltd | AnalystLab Africa Week 1, Data Analytics Track | Divine Okezie Nwaoha

1. Assumptions and definitions

Item	Definition used
Which orders count	Completed orders only (16,529). Cancelled and Pending orders are excluded from all revenue, order, quantity and return figures.
Revenue	Quantity × Unit_Price − Discount (the Line_Total in Order_Items). Discount is a flat naira amount, not a percentage. Shipping cost is not included.
Average Order Value	Total Revenue ÷ number of Completed orders (₦151,057.20).
Average Selling Price	Average Unit_Price on Completed order lines (₦50,364.42).
Total Customers	All 5,000 registered customers in the Customers table, not only those who bought.
Return Rate	Completed orders that were returned ÷ Completed orders = 798 ÷ 16,529 = 4.83%.
Low stock	A product is low stock when Stock_On_Hand is at or below its Reorder_Level.
Inventory Value	Stock_On_Hand × Cost_Price, summed across all inventory rows = ₦3,169,020,818.
Membership Level	The Customer_Segment column (Bronze, Silver, Gold).
Region	The region of the store where the sale happened (from the Stores table).
Store identity	Store names repeat across stores (for example ShopSmart Abuja is STR001, STR010, STR017 and STR023), so stores are shown as Store_ID plus name (Store Label).
Supplier link	Products have no Supplier_ID. Each supplier name is the product brand plus " Supplies", so products were matched to suppliers on that pattern.

2. Limitations

- **Currency symbol.** Workbook cells display $, but all amounts are Nigerian naira (₦), as in the source data. Titles and labels in reports use ₦.
- **Customers grouped by name.** The Top 10 Customers pivot uses customer names, not Customer_ID, so customers who share a name could be combined into one row.
- **Pivots not sorted.** Pivots, including Top 10 and Bottom 10 Products, are in default order. Ranking is left to the dashboard stage.
- **Ties in returns.** Eight of the ten most returned products have 5 returns each, so the tenth place is a tie and other products may share it.
- **One year of data.** All orders fall in 2025, so trends and seasonality (such as the October to December peak) are based on a single year.
- **Supplier link is inferred.** It rests on a naming pattern, not an official key, so it should be confirmed with the client.
- **Fixed copy of order data.** The Sales table was pasted as values and its rows follow the same order as the Orders sheet. Sorting or editing Orders later would not update it.
- **Dashboard filters and inventory.** Inventory Value and Total Customers come from separate tables, so dashboard slicers built on the Sales table do not filter them.
- **Phone numbers.** Formats are inconsistent (some include extensions marked with "x"). They were left unchanged because no analysis uses them.

3. Corrections made during the analysis

Issue found	Correction
Order status lookup stopped at row 10,000, so many order lines showed "Not found" and revenue was understated.	Lookup range extended to cover every order. Total revenue rose to ₦2,496,824,383 and Total Quantity to 49,432.
Number of Customers showed 9,999, the size of the formula range.	Corrected to 5,000 (confirmed by selecting the Customer_ID column).
Supplier lookup returned errors because brand names did not match supplier names exactly.	Matched on Brand + " Supplies", which gave a match for the products.
Return flag formula referred to its own cell (circular reference) and returned zeros.	Rebuilt to read Order_ID from the Orders sheet and count matches in Returns.
The Sales table had no Order_ID, which the return check needs.	Order_ID added by position from the Orders sheet, then pasted as values.

Issue found	Correction
Return Rate by Category pivot showed rates above 100% (formula reversed, shown as % of grand total).	Calculated field corrected to Completed_Returned ÷ Completed_Order, shown with no calculation. Grand total now 4.83%.
Returns sheet has 2,500 records, but 2,442 orders were flagged as returned.	The difference of 58 is attributed to orders that appear more than once in Returns. This was not separately confirmed.
Regional answer needed more context.	South leads on total revenue because it has 8 of 25 stores; revenue per store is similar in every region (₦97M to ₦101M). The report states both.